

MEDIATEK

MT8676 Yocto USB User Manual

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Version History

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1 MTK USB

1.1 Overview

Both the Host and Device can support up to USB 3.1 Gen1 (5GHz), with BC1.2 when a USB 2.0 Device is used.

- Host xHCI supports USB 3.0 Physical Port, USB 2.0 Physical Port and up to 32 EPs.
- Device MTU3 supports 8 TX/RX EPs with 8K SRAM, and there are 8 USB MTK Classic Accessory Channel queues.

1.2 MTK USB Architecture Relation Overview

Figure 1-1 shows the block diagram of MTK USB in Demo Board.

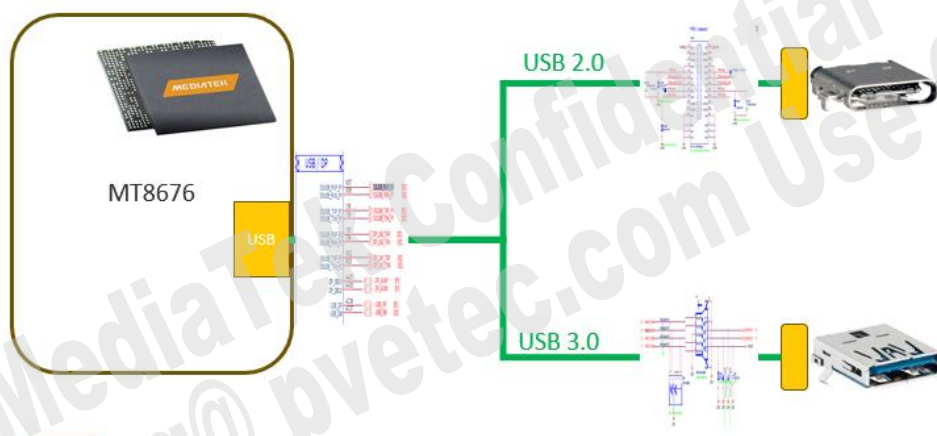


Figure 1-1. MT8676 MTK USB HW diagram

MT8676 has a unique MTK USB IP. The MTK SLT Demo Board named MT8676_P1V1_SMT2 is used to split USB 2.0 and USB 3.1:

- Wired Type-C Port: Plans to make it connected with Desktop, and it acts as Device for ADB debug usage.
- Wired Type-A Port: Plans to make it support the Open USB Jack, and it acts as Host for any USB device.

Anyway, we have to just keep one port wired in Type-C Jack or Type-A Jack for logical connection well, since there is not designed in switcher control to make one port wired in Type-C Jack or Type-A Jack for logical connection automatically. Keep physical connection on one port so that it could obey USBIF compliance requirement.

- Type-C Port:

USB2.0 Port is plugging in USB line wired with Desktop, make it default Device mode, and concern USB-A port without any USB device. SMT1 does not support Type-C yet in SMT1 Demo board.

- Type-A Port:

USB3.1 Gen1 Port is plugging in any USB device with USB 3.0 or lower version, it fully supports the USB 3.1 Gen1 spec. The following CLI commands should be used to switch it to Host mode.

Keep Power Save active.

```
su
echo lock > /sys/power/wake_lock
```

Host mode:

```
echo 2 >/sys/devices/platform/soc/11201000.usb0/mode
```

Device mode :

```
echo 3 >/sys/devices/platform/soc/11201000.usb0/mode
```

None mode:

```
echo 0 >/sys/devices/platform/soc/11201000.usb0/mode
```

Note:

- To recommend using these sequences to change USB Role: ...-->None-> Host->None---Device---None->...

1.3 MTK USB Kernel Configuration

- Location of MTU3:

```
<work project>/kernel/kernel_device_modules-6.1/drivers/usb/mtu3
```

- The Kernel of MTU3 should be configured as:

```
CONFIG_DEVICE_MODULES_USB_MTU3=m
```

```
CONFIG_DEVICE_MODULES_USB_MTU3_DUAL_ROLE=y
```

- Property list of MTU3:

Note: For property that is not listed here, it is not recommended for you to change it except that you are an expert.

- mediatek,u3p-dis-msk: each 1 bit means one port, with the LSB bit0 representing Port1, and so on.
- enable-nanual-drd: It is seldom used since it is only applicable in some scenarios without USB role or extcon.
- usb-role-switch: It is mainly used since it is perfect for the upstream.
- extcon: It only can be used since the upstream prefers not to use this style.

- Location of xHCI:

```
<work project>/kernel/kernel_device_modules-6.1/drivers/misc/mediatek/usb_xhci
```

- The Kernel of xHCI should be configured as:

```
CONFIG_DEVICE_MODULES_USB_XHCI_MTK=m
```

- Property list of xHCI:

Note: For property that is not listed here, it is not recommended for you to change it except that you are an expert.

- usb3-lpm-capable: Each 1 bit represents one port, with the LSB bit0 representing Port1, and so on.

If the parent node is MTU3, this property is ignored.

- Location of xSPHY:

```
<work project>/kernel/kernel_device_modules-6.1/drivers/phy/mediatek
```

- The Kernel of xSPHY should be configured as:

CONFIG_DEVICE_MODULES_PHY_MTK_XSPHY=m

- Property list of xSPHY:

Note: For property that is not listed here, it is not recommended for you to change it except that you are an expert.

- For the parameters of the device eye pattern:

- mediatek,eye-src
- mediatek,eye-vrt
- mediatek,eye-term
- mediatek,rev6

- Regarding the host eye-pattern parameters:

- mediatek,eye-src-host
- mediatek,eye-vrt-hos
- mediatek,eye-term-host
- mediatek,rev6-host
- mediatek,disctb

1.3.1 MTK USB Connector Status

SMT1															
Port	TYPE A		OTG			TYPE C (RTQ7883)								Jack	Comment
						I2C (5th)		vBus CTRL		Switcher		Enable	Intr		
	vBus Ctrl	vBus Fault	ID	vBus Det	vBus CTRL	SC L	SD A	PMI C	GPIO	Mux Power	Mux Sel	Enable	Type C Intr		
Port0	Yes					Yes									Choose One in Type-C and Type-A
				6		163	164					98		Type C	Default Device with USB 2.0, and Manual switch USB Role
										101	102			Type A	Default Device with USB 2.0, and Manual switch USB Role
														Micro USB	
SMT2															
Port	TYPE A		OTG			TYPE C (RTQ7883)								Jack	Comment
						I2C (5th)		vBus CTRL		Switcher		Enable	Type C Intr		
	vBus Ctrl	vBus Fault	ID	vBus Det	vBus CTRL	SC L	SD A	PMI C	GPIO	Mux Power	Mux Sel	Enable	Type C Intr		

Po rt0	Yes				Yes								Choose One in Type-C and Type-A	
			6		1 6 3	1 6 4					98	48	Type C	
										101	102		Type A	Obey Type-C Role
													Micro USB	

1.3.2 MTK USB Kernel General USB Connector Configuration

- Location of USB CONN:

```
<work project>/kernel/kernel_device_modules-6.1/drivers/misc/mediatek/extcon
```

- The Kernel of USB CONN should be configured as:

```
CONFIG_DEVICE_MODULES_USB_CONN_GPIO=m
```

- Property list of USB CONN:

One of id-gpio or vbus-gpio must be present, and both can also be present.

- id-gpio: GPIO for the USB ID pin. See GPIO binding.
- vbus-gpio: GPIO for the USB VBUS pin.
- Wakeup source: GPIO used for wake source.

1.3.3 MTK USB USBIF Compatibility Patch to Cherry-pick

These patches need to be cherry-picked to your local project. You can proceed with the process.

The customer project manager is assigned to them.

https://gerrit.mediatek.inc/c/quark/kernel_device_modules-6.1/+8792849

https://gerrit.mediatek.inc/c/quark/kernel_device_modules-6.1/+8931041

https://gerrit.mediatek.inc/c/quark/kernel_device_modules-6.1/+8931042

```
cd <work project>/kernel/kernel_device_modules-6.1/
git apply <braaa.patch>
```

- Location of USBIF compatibility patch:

```
<work project>/kernel/kernel_device_modules-6.1/drivers/misc/mediatek/usb_xhci
<work project>/kernel/kernel_device_modules-6.1/drivers/usb/mtu3
<work project>/kernel/kernel_device_modules-6.1/drivers/phy/mediatek
```

- USBIF compatibility patch should be configured as:

```
CONFIG_DEVICE_MODULES_USB_MTK_HQA_TEST=y
```

Notes:

- To locate the position of 'hqa' in Host mode, use the following command: `find /sys/devices -name hqa -type f`.
- The 'lsusb' command should display the USB devices that are plugged in, helping you to confirm whether it is in host mode or not.

1.4 MT8676 USBIF Compliance Program for xHCI USB 2.0

1.4.1 Some Background Information

MT8676 can support 1 USB 2.0 High-Speed port, and Phy register layout information is provided.

port	offset	bank
u2 port0	0x0000	MISC
	0x0100	FMREG
	0x0300	U2PHY_COM
u3 port0	0x0700	SPLL
	0x0800	CHIP
	0x0900	U3PHYD
	0x0a00	U3PHYD_BANK2
	0x0b00	U3PHYA
u2 port1	0x0c00	U3PHYA_DA
	0x1000	MISC
	0x1100	FMREG
	0x1300	U2PHY_COM
u3 port1	0x1700	SPLL
	0x1800	CHIP
	0x1900	U3PHYD
	0x1a00	U3PHYD_BANK2
	0x1b00	U3PHYA
u2 port2	0x1c00	U3PHYA_DA
	0x2000	MISC
	0x2100	FMREG
	0x2300	U2PHY_COM

port	offset	bank
The MISC value for u2 port0 is 0x0000.		
	0x0100	FMREG
	0x0300	U2PHY_COM
u3 port0	0x0700	SPLL
	0x0800	CHIP
	0x0900	U3PHYD
	0x0a00	U3PHYD_BANK2
	0x0b00	U3PHYA
	0x0c00	U3PHYA_DA
	U2 port1: 0x1000	MISC
	0x1100	FMREG
	0x1300	U2PHY_COM
The u3 port1 is configured with the value 0x1700 for the SPLL.		

```

0x1800    CHIP
0x1900    U3PHYD
0x1a00    U3PHYD_BANK2
0x1b00    U3PHYA
0x1c00    U3PHYA_DA

```

The u2 port2 has an address of 0x2000 in the MISC.

```

0x2100    FMREG
0x2300    U2PHY_COM

```

1.4.2 How to Enable xHCI USB 2.0 Toolkits?

CONFIG_DEVICE_MODULES_USB_MTK_HQA_TEST=y

CONFIG_DEBUG_FS=y

1.5 Parameters to Tune xHCI USB 2.0 Eye-pattern Quality

1.5.1 Tune Phy RG_USB20_VRT_VREF_SEL Parameters

- VRT reference voltage selection
 - Register:

11C21304		USBPHYACR1		U2PHYA Common 1 Register								00A44400					
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	
Name									RG_USB20_INTR_CAL					RG_USB20_OTG_VBUSTH			
Type									RW					RW			
Reset									1	0	1	0	0	1	0	0	
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0	
Name		RG_USB20_VRT_VREF_SEL				RG_USB20_TERM_VREF_SEL				RG_USB20_MPX_SEL							
Type		RW				RW				RW							
Reset		1	0	0		1	0	0	0	0	0	0	0	0	0	0	

- Bit field:

14:12 RG_USB20_VRT_VREF_SEL

VRT reference voltage selection(share circuit)

```

000:700mV
001:720mV
010:740mV
011:760mV
100:770mV
101:780mV
110:800mV
111:820mV

```

1.5.2 Tune Phy RG_USB20_TERM_VREF_SEL Parameter

- HS_TX TERM reference voltage selection
 - Register:

11C21304	USBPHYACR1				U2PHYA Common 1 Register								00A44400			
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name									RG_USB20_INTR_CAL				RG_USB20_OTG_VBUSTH			
Type									RW				RW			
Reset									1	0	1	0	0	1	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RG_USB20_VRT_VREF_SEL					RG_USB20_TERM_VREF_SEL				RG_USB20_MPX_SEL						
Type	RW					RW				RW						
Reset		1	0	0		1	0	0	0	0	0	0	0	0	0	0

– Bit field:

10:8 RG_USB20_TERM_VREF_SEL

HS_TX TERM reference voltage selection(share circuit)

000:320mV

001:340mV

010:360mV

011:380mV

100:400mV

101:420mV

110:440mV

111:460mV

1.5.3 Tune PHY RG_USB20_HSTX_SRCCTRL Parameter

- High speed slew rate control

– Register:

11C21314	USBPHYACR5				U2PHYA Common 5 Register								00801000			
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	u2phy_reserve14				RG_USB20_DISC_FIT_EN				RG_USB20_INIT_SQ_EN_DG				RG_USB20_HSTX_TMODE_SEL			
Type	RO				RW				RW				RW			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name	RG_USB20_HSTX_SRCAL_EN				RG_USB20_HSTX_SRCCTRL				RG_USB20_HS_100U_U3_EN				RG_USB20_GBIAS_ENB			
Type	RW				RW				RW				RW			
Reset	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

– Bit field:

14:12 RG_USB20_HSTX_SRCCTRL

high speed slew rate control

000: min rise time/fall time

111: max rise time/fall time

1.5.4 Tune PHY RG_USB20_PHY_REV Parameter

- High speed pre-emphasis control

– Register:

11C20318	USBPHYACR6				U2PHYA Common 6 Register								00900482			
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name	RG_USB20_PHY_REV												RG_USB20_BC11_SW_EN			
Type	RW				RW								RW			
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name					RG_USB20_HSRX_MMODE_SEL				RG_USB20_HSRX_BIAS_EN_SEL				RG_USB20_HSRX_TMODE_EN			
Type					RW				RW				RW			
Reset					0	0	0	0	1	0	0	0	0	0	0	0

– Bit field:

31:30 RG_USB20_PHY_REV: pre-emphasis control

2b'00 means no drive

2b'01 means 1st gear

2b'10 means 2nd gear

2b'11 means 3rd gear

1.6 How to Tune USBIF Eye-pattern Parameter in xHCI?

1.6.1 How to Locate Position of hqa?

```
su
dmesg -n 1
```

Find files named 'hqa' under the '/sys' directory.

You can now view the following echo statement below.

```
/sys/devices/platform/soc/11201000.usb0/11200000.xhci0/hqa
/sys/devices/platform/soc/11201000.usb0/11200000.xhci0 is location of hqa.
```

1.6.2 Change hqa Directory as Current Work Directory

```
cd /sys/devices/platform/soc/11201000.usb0/11200000.xhci0
```

```
ls -al
```

You should watch these hqa and RG* files relations below.

1.6.3 Some Files for Params of Eye-pattern to Tune

```
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_CHGDT_EN
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_BGR_DIV
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_DISCTH
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_HSTX_SRCTRL
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_INTR_EN
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_PHY_REV
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_PLL_BW
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_REV4
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_SQTH
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_TERM_VREF_SEL
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 RG_USB20_VRT_VREF_SEL
```

1.6.4 USBIF USB 2.0 Compliance Toolkit

```
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 hqa
```

1.6.5 Force USB 3.1 Gen1 Compliance Mode for USB 3.0 Compliance

```
-rw-r--r-- 1 root root 4096 2024-03-02 09:11 usb3hqa
```

1.7 How to Understand u2p, Index Parameters for CLI: RG* or hqa

1.7.1 hqa

To obtain online help, type `cat hqa` to access a wealth of useful information.

```
echo -n item port-id > hqa
```

General format: USB<spec:20 or 30> Port<port-id: n>

The chosen item should be one of the following statements:

test.j: Test_J

test.k: Test_K

test.se0: Test_SE0_NAK

test.packet: Test_PACKET

test.suspend: Port Suspend

test.resume: Port Resume

test.enumbus: Enumerate Bus

test.getdesc: Get Device Descriptor

test.debug: Debug port information

pm.u1u2: Port U1,U2

Select a suitable port ID number from the options provided in the online help.

USB30 Port1: 0x0A0003C0

The USB30 Port1 indicates that it complies with the USB 3.1 Gen1 specification, with a port ID of 1.

USB20 Port2: 0x00000E03

USB20 Port2 PORTMSC[31,28] 4b'0000: 0x00000000

The USB20 Port2 indicates that it follows the USB 2.0 specification, with a port ID of 2.

PORTMSC[31,28] is debug information feedback, bitmap means TEST_J, TEST_K, TEST_SE0_NAK and TEST_PACKET for compliance analog test item.

You have to reboot DUT to clean PORTMSC[31,28] before manipulating next MAC compliance experiment.

Note:

test.debug is not special for compliance design, while it is special for debug which port to expect. It is useful when you don't understand mapping relation between real port and port to expect.

The test item information is utilized for exporting log information through the use of printk.

The HQA relation information can be obtained using a CLI command.

```
dmesg |grep xhci
```

Example: Plug in one USB device into the expected port, and use the following commands:

```
console:/sys/devices/platform/soc/11201000.usb0/11200000.xhci0 # cat hqa
info:
    echo -n item port-id > hqa
port-id: based on the number of USB3 ports, e.g.
        xHCI with 1 u3p, 2 u2p: 1st u2p-id is 2(1+1), 2nd is 3
items:
    test.j: Test_J
    test.k: Test_K
    test.se0: Test_SE0_NAK
    test.packet: Test_PACKET
    test.suspend: Port Suspend
    test.resume: Port Resume
    test.enumbus: Enumerate Bus
test.getdesc: Get Device Descriptor
test.debug: debug Port information
    pm.ulul2: Port U1,U2
USB30 Port1: 0x00001203
Powered, connected, and enabled. Link: U0. PortSpeed: SuperSpeed Gen1x1. Change: Wake:
USB20 Port2: 0x0A0002A0
Powered, not connected, disabled link: RxDetect PortSpeed: UNKNOWN-Speed Change: Wake: WCE
WOE
USB20 Port2 PORTMSC[31,28] 4b'0000: 0x00000000
console:/sys/devices/platform/soc/11201000.usb0/11200000.xhci0 # echo -n test.debug 1 >hqa
console:/sys/devices/platform/soc/11201000.usb0/11200000.xhci0 # dmesg |grep xhci
[ 612.198919] sh: xhci-mtk 11200000.xhci0: [name:xhci_mtk_hcd_v2&][0] test.debug
[ 612.198966] sh: xhci-mtk 11200000.xhci0: [name:xhci_mtk_hcd_v2&][1] 1
[ 612.198990] sh: xhci-mtk 11200000.xhci0: [name:xhci_mtk_hcd_v2&]mu3h t_debug_port test
port1
[ 612.199016] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]configured
[ 612.199035] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]New USB device found, idVendor=13fe,
idProduct=6300, bcdDevice= 1.00
[ 612.199059] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]New USB device strings: Mfr=1, Product=2,
SerialNumber=3
[ 612.199078] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]Product: USB DISK 3.0
[ 612.199096] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]Manufacturer:
[ 612.199111] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]SerialNumber: 0700378539191802
[ 612.199128] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]D: Ver= 3.20 Cls=00(>ifc ) Sub=00
Prot=00 MxPS= 9 #Cfgs= 1
[ 612.199155] sh: usb 2-1: [name:xhci_mtk_hcd_v2&]P: Vendor=13fe ProdID=6300 Rev= 1.00
```

1.7.2 RG*

Get online help by typing `cat RG_USB20_TERM_VREF_SEL`, then you can get many useful information.

`echo -n u2p index binary_format_value > RG_USB20_TERM_VREF_SEL`

General format: USB<spec:20 or 30> Port<u2p: n> (Phy<index: m> enable)

```
USB20 Port2: 0x00000E03
USB 2.0 Port 2 (Phy0: enabled): 0x11E40000 0x000004004
  RG_CHGDT_EN           = 1b0
  RG_USB20_BGR_DIV      = 2b10
  RG_USB20_DISCTH       = 4b1001
  RG_USB20_HSTX_SRCTRL  = 3b011
  RG_USB20_INTR_EN      = 1b1
  RG_USB20_PHY_REV      = 2b01
  RG_USB20_PLL_BW       = 3b011
  RG_USB20_REV4         = 1b0
  RG_USB20_SQTH         = 4b0010
  RG_USB20_TERM_VREF_SEL = 3b100
  RG_USB20_VRT_VREF_SEL = 3b100
```

USB20 Port2 (Phy0: enabled) means it's USB20 that follows the USB 2.0 spec, and its u2p is 2; its index is 0.

The following is a Demo:

For USB 2.0, tune this `RG_USB20_VRT_VREF_SEL = 3b100` less, then `cat RG_USB20_VRT_VREF_SEL`.

```
Usage: current HQA setting check
USB30 Port1: 0x0A0002A0
USB20 Port2: 0x0A0002A0
USB20 Port2 (Phy0: enable): 0x11E40000 0x000004004
  RG_CHGDT_EN           = 1b0
  RG_USB20_BGR_DIV      = 2b10
  RG_USB20_DISCTH       = 4b1001
  RG_USB20_HSTX_SRCTRL  = 3b011
  RG_USB20_INTR_EN      = 1b1
  RG_USB20_PHY_REV      = 2b01
  RG_USB20_PLL_BW       = 3b011
  RG_USB20_REV4         = 1b0
  RG_USB20_SQTH         = 4b0010
  RG_USB20_TERM_VREF_SEL = 3b100
  RG_USB20_VRT_VREF_SEL = 3b100
```

RG_USB20_VRT_VREF_SEL usage:

```
echo u2p index 3b011 > RG_USB20_VRT_VREF_SEL
parameter: u2p: 2
parameter: index: 0
e.g.: echo 2 0 3b101 > RG_USB20_VRT_VREF_SEL
port2 binding phy 0, tune 3b'010 as VRT_VREF value
```

Note:

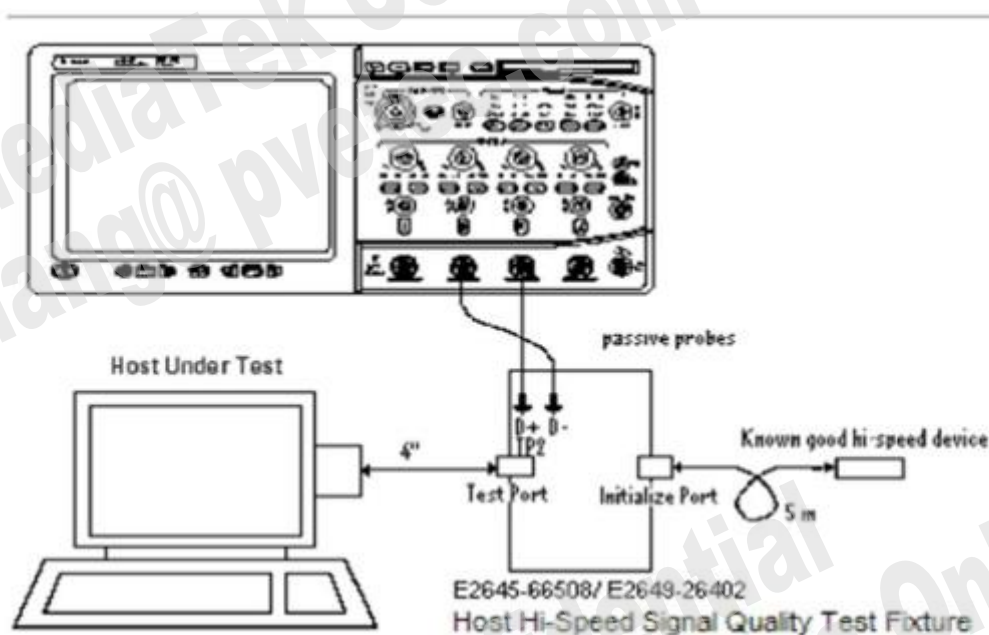
14:12	RG_USB20_VRT_VREF_SEL	VRT reference voltage selection (shared circuit): 000: 700mV 001: 720mV 010: 740mV 011: 760mV 100: 770mV 101: 780mV 110: 800mV 111: 820mV
-------	-----------------------	---

1.7.3 Don't Forget to Send USB Engineer These Suitable Value of These Parameter Registers Tied in Eye-pattern Report

Submit your parameters tied in eye-pattern to *.dts file of your project.

```
&u2port0 {
mediatek,eye-src-host = <0x04>;
mediatek,eye-vrt-host = <0x04>;
mediatek,eye-term-host = <0x07>;
mediatek,rev6-host = <0x07>;
status = "okay"
}
```

1.8 How to Tune Eye-pattern Parameters in Device Mode USBIF Compliance Test



Toolkit: <https://www.usb.org/document-library/xhsett>

Manual Guide: https://www.usb.org/sites/default/files/HSETT_Instruction_0_4_1.pdf

1.8.1 Eye-pattern Parameters Location in Device Mode

```
console:/proc/mtk_usb/usb-phy0/u2_phy # ls -al
total 0
```

```
dr-xr-xr-x 8 root root 0 2024-03-28 13:20 .
dr-xr-xr-x 4 root root 0 2024-03-28 13:20 ..
-rw-r--r-- 1 root root 0 2024-03-28 13:20 discth
-rw-r--r-- 1 root root 0 2024-03-28 13:20 intr_ofs
-rw-r--r-- 1 root root 0 2024-03-28 13:20 phy_rev6
-rw-r--r-- 1 root root 0 2024-03-28 13:20 rx_sqth
-rw-r--r-- 1 root root 0 2024-03-28 13:20 term_sel
-rw-r--r-- 1 root root 0 2024-03-28 13:20 vrt_sel
```

1.8.2 Eye-pattern Parameters Value in Device Mode

```
console:/proc/mtk_usb/usb-phy0/u2_phy # cat *
discth = 1001
intr_ofs = 0
intr_val = 100011
phy_rev6 = 01
rx_sqth = 0010
term_sel = 100
vrt_sel = 010
```

1.8.3 Change Eye-pattern Parameters Setting in Device Mode

The values should be changed to the '0' prefixed binary system format.

For example, if discth = 1001, it means its value is in base 9. To convert it to base 10, follow the step below:

```
echo 01010 > discth
```

1.8.4 Don't Forget to Send USB Engineer These Suitable Value of These Parameter Registers Tied in Eye-pattern Report

Submit your parameters tied in eye-pattern to *.dts file of your project:

```
&u2port0 {
mediatek,eye-src = <0x04>;
mediatek,eye-vrt = <0x04>;
mediatek,eye-term = <0x07>;
mediatek,rev6 = <0x07>;
status = "okay"
}
```

1.9 How to Understand *usb3hqa* Parameter for CLI: *usb3hqa* for USB 3.1 Gen1 USBIF Compliance Test

1.9.1 *usb3hqa*

1.9.1.1 Recommend to Use Human Easy Commands to Trigger Compliance Mode

Online help can be obtained by typing `cat usb3hqa`, which will provide a wealth of useful information.

```
echo -n item port-id > usb3hqa
```

General format: USB<spec:30> Port<port-id: n>

Type the following command in the CLI: `cat usb3hqa`

Select a suitable port ID number from the options provided in the online help.

USB30 Port1: 0x0A0003C0

The *USB30 Port1* indicates that it adheres to the USB 3.1 Gen1 specification, with a port ID of 1.

PORTPLS[8,5] is debug information feedback, bitmap means U0, U1, U2 and so on, here 0x0A means “Compliance mode” for analog compliance testing.

Note:

- Type `cli: cat usb3hqa | grep Compliance` to confirm whether it is in Compliance mode.

1.9.1.2 Force RG Enter USB 3.1 Gen1 USBIF Compliance Mode in Expert Manner

```
console:/sys/devices/platform/soc/11201000.usb0/11200000.xhci0 # cat reg
SSUSB register operation interface help info.
rx - read xhci reg: offset [len]
rm - read mu3d reg: offset [len]
ri - read ippc reg: offset [len]
rp - read phy3 reg: offset [len]
ru - read phy2 reg: offset [len]
wx - write xhci reg: offset value
wm - write mu3d reg: offset value
wi - write ippc reg: offset value
wp - write phy3 reg: offset value
wu - write phy2 reg: offset value
sx - set xhci mac reg bits: offset bit_start mask value
sm - set mu3d mac reg bits: offset bit_start mask value
si - set ippc reg bits: offset bit_start mask value
sp - set phy3 reg bits: offset bit_start mask value
su - set phy2 reg bits: offset bit_start mask value
px - print xhci mac reg bits: offset bit_start mask
pm - print mu3d mac reg bits: offset bit_start mask
```

```

pi - print ippc      reg bits: offset bit_start mask
pp - print phy3      reg bits: offset bit_start mask
pu - print phy2      reg bits: offset bit_start mask

```

Note:

- Numbers should be HEX, except bit_star (DEC)

echo wx 0x420 0x10340 > reg

1.10 xHCI USB 2.0 USBIF Compliance Test

1.10.1 Host High-speed Signal Quality (EL_2, EL_3, EL_6, EL_7)

CLI: echo -n test.packet [port] > hqa

1.10.2 Host Controller Packet Parameter (EL_21, EL_22, EL_23, EL_25, EL_55)

CLI: echo -n test.getdesc [port] > hqa

1.10.3 Host CHIRP Timing (EL_33, EL_34, EL_35)

CLI: echo -n test.enumbus [port] > hqa

1.10.4 Host Suspend/Resume timing (EL_39, EL_41)

CLI (suspend): echo -n test.suspend [port] > hqa

CLI (resume): echo -n test.resume [port] > hqa

1.10.5 Host Test J/K, SE0_NAK (EL_8, EL_9)

CLI (J): echo -n test.j [port] > hqa

CLI (K): echo -n test.k [port] > hqa

CLI (SE0): echo -n test.se0 [port] > hqa

1.10.6 Drop Test

CLI: none

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